

Renewable Energy Generation Dashboard (India) State-wise Analysis (2020–2025)

Project Documentation

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Project Overview

This project analyzes renewable energy generation across Indian states using Excel, Power Query and Power BI. The objective is to clean, transform and visualize renewable energy data from 2020–2025 to identify trends, regional performance and dominant renewable energy sources

Objectives

- Analyze renewable energy generation by state and region.
- Compare wind, solar and other renewable sources.
- Study yearly trends.
- Build an interactive Power BI dashboard.

Data Source

- India Data Portal
- Central Electricity Authority (CEA)
- Renewable Energy Generation Dataset (2020–2025)
- Link: <https://indiadataportal.com/p/power/r/mop-renewable-energy-state-st-dl-abc>

Data Cleaning & Transformation

- Removed duplicate records.
- Checked and handled missing values.
- Standardized column formats.
- Converted Date column to Date data type.

- Created Year and Month columns.
- Added Month Number column.
- Created Energy Category.
- Created Dominant Energy Source.
- Validated numerical columns.
- Built Pivot Tables for summary analysis.

Dataset Columns

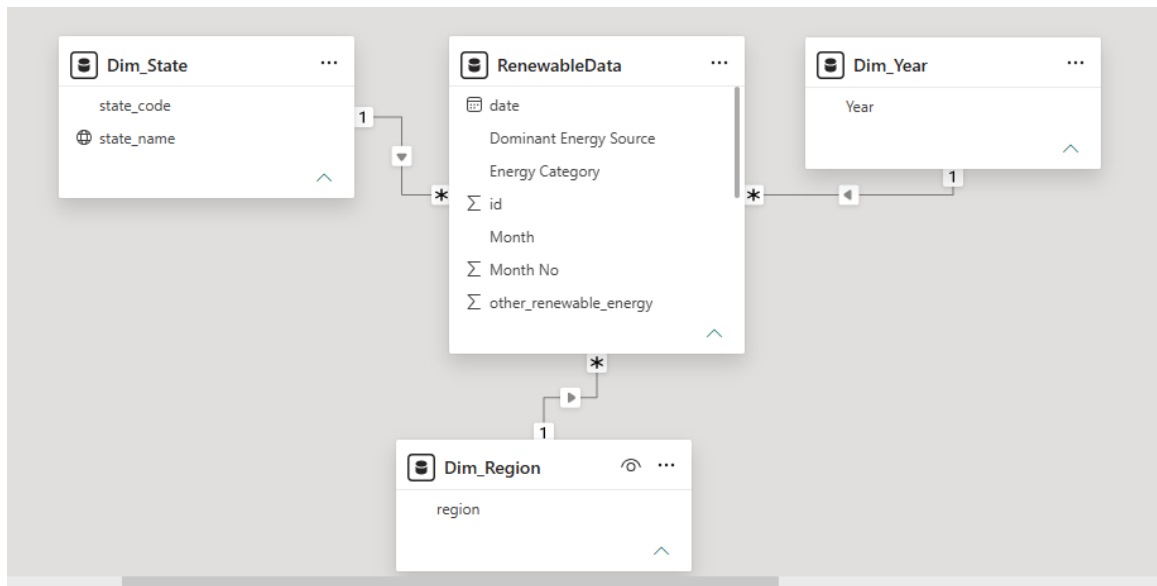
Column	Type	Description
id	Integer	Unique ID
date	Date	Date
state_name	Text	State
region	Text	Region
wind_energy	Decimal	Wind
solar_energy	Decimal	Solar
other_renewable_energy	Decimal	Other
total_renewable_energy	Decimal	Total
Year	Integer	Year
Month	Text	Month
Energy Category	Text	Category
Dominant Energy Source	Text	Dominant

Tools & Technologies

- Excel: Data cleaning, transformation, and Pivot Tables.
- Power BI: Data modelling, DAX calculations, visualization, and interactive dashboard creation.

Data Model

Star Schema was implemented.



Fact Table

- Renewable Data

Dimension Tables

- Dim State
- Dim Region
- Dim Year

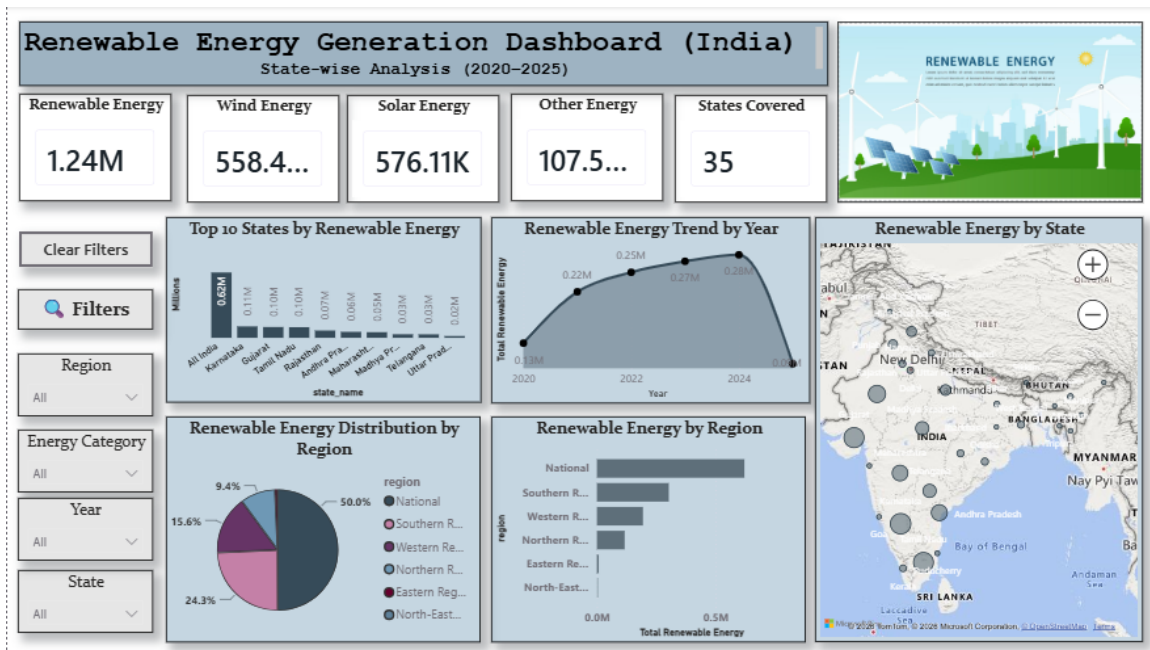
Relationship Type

- One-to-Many (1:*)

Dashboard Features

- KPI Cards
- Interactive Filters
- Top 10 States Analysis
- Yearly Trend Analysis
- Regional Distribution
- Geographic Map
- Pie Chart
- Dynamic Filtering

Dashboard



DAX Measures

- Total Renewable Energy
- Total Wind Energy
- Total Solar Energy
- Total Other Renewable Energy
- Average Renewable Energy
- Maximum Renewable Energy
- Minimum Renewable Energy
- Total States

Key Insights

- Total renewable energy generation exceeded **1.24M units**.
- Wind and Solar energy contribute the largest share of renewable generation.
- National Region recorded the highest renewable energy generation.
- Renewable energy production showed consistent growth between 2020 and 2024.
- Interactive filters allow users to analyze data by Region, State, Year, and Energy Category.

Conclusion

This project demonstrates a complete Business Intelligence workflow, beginning with data cleaning and transformation in Excel and Power Query, followed by data modeling, DAX calculations, and interactive dashboard development in Power BI. The dashboard enables users to analyze renewable energy generation across Indian states efficiently and supports data-driven decision-making.